

An Integrated OCR and NLP-Based System for Automated Classification and Information Extraction of Church Administrative Documents

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Abstract—The manual management of administrative documents in community-based organizations like churches, encompassing items such as permit letters, invitations, and loan agreements, is often inefficient and prone to human error. This research proposes an integrated system that leverages Optical Character Recognition (OCR) and Natural Language Processing (NLP) to automate the classification and information extraction from these documents. The system utilizes PyTesseract for robust text extraction from PDF and image files, and a fine-tuned IndoBERT model for both document classification and Named Entity Recognition (NER). Experimental results demonstrate high performance, with the classification model achieving an accuracy of 93.6% on new data and the NER model reaching an F1-Score of 0.989. Furthermore, user acceptance testing showed a significant improvement in efficiency, reducing document processing time by up to 69.6%, and achieved a System Usability Scale (SUS) score of 77.5, indicating good usability. This study confirms that the proposed system is a functional, efficient, and well-received solution for enhancing the accuracy and efficiency of church administrative document management.

Keywords—Document Management, Optical Character Recognition (OCR), Natural Language Processing (NLP), Named Entity Recognition (NER), BERT, Information Extraction, Document Classification.

I. INTRODUCTION

The management of administrative documents in many community-based organizations, such as the St. John the Apostle Chapel, remains a largely manual process. Staff members must individually read, categorize, and record information from various documents like permit requests, event invitations, and asset loan forms. This traditional workflow is not only time-consuming but also susceptible to human errors, including incorrect data entry and the misplacement of physical documents. These documents often exhibit semi-structured formats, combining formal letterheads with varied layouts and domain-specific terminology related to liturgical events and church assets, which poses unique challenges for generic document processing systems. While Optical Character Recognition (OCR) tools exist to digitize documents, they often lack the integrated intelligence to understand and structure the extracted text, leaving a significant gap in automation.

To address these challenges, this paper presents the design and implementation of an integrated system for automated document classification and information extraction. Our system combines OCR technology to convert unstructured documents (PDFs and images) into machine-readable text, with a sophisticated Natural Language Processing (NLP)

model to analyze the content. Specifically, we utilize a Bidirectional Encoder Representations from Transformers (BERT) model, fine-tuned for the Indonesian language (IndoBERT), to perform two critical tasks simultaneously. This model was specifically chosen as it has been pre-trained on a large Indonesian corpus and has demonstrated state-of-the-art performance, outperforming multilingual models on various Indonesian NLP tasks [1]. Its selection is crucial for accurately understanding the contextual nuances and specific terminology present in the administrative documents of the target institution. The model classifies documents into predefined categories and extracts key entities such as dates, locations, event names, and loan items.

The primary contribution of this work is a functional and efficient end-to-end system that significantly reduces manual effort and improves data accuracy in administrative document management, particularly within a domain that has received limited attention in automated processing research. The proposed model serves as a practical solution for the target institution and offers a framework that can be adapted by other organizations facing similar challenges. Our work is grounded in established principles of NLP [2] and leverages state-of-the-art transformer architectures to solve a real-world problem. The integration of OCR and NLP has proven effective not only in administrative tasks but also in diverse fields such as invoice processing [3], medical prescription analysis [4], and automated resume screening [5], demonstrating its robustness and versatility.

II. RELATED WORKS

The task of automatically processing documents draws upon advances in both OCR and NLP. OCR technology, particularly open-source engines like Tesseract, has been widely implemented for text extraction from images [6], [7]. However, the raw text output from OCR requires further semantic analysis to be truly useful. This is where NLP, and specifically Named Entity Recognition (NER), plays a crucial role. NER aims to identify and categorize key information entities within text [8], a task essential for document understanding.

Several studies have explored the integration of OCR and NER for various applications. For example, Najmi & El-Dosuky [9] demonstrated a system for extracting data from confidential documents, while Khairani et al. [10] applied this combination to detect ingredients in food packaging for the Indonesian market, highlighting its relevance within the IEEE community. Other works have shown success in specific domains, such as processing job application forms [11] and automating the review of construction specifications [12]. The

versatility of this integrated approach is further highlighted in other specialized domains. For instance, Kulkarni et al. [3] developed a framework using Google OCR and a BERT-enabled Deep Convolutional Neural Network (DCNN) to structure and classify invoice documents, achieving an accuracy of 94.7%. Similarly, Sakib et al. [4] applied a combination of computer vision (VGG-16) and a BERT model to extract and classify text from medical prescriptions, underscoring the technology's value in the healthcare sector. The broader industrial relevance is also evident in systems for automated resume analysis, which leverage OCR and NLP to extract key information and match candidates with job descriptions, significantly improving screening efficiency [5].

While the aforementioned studies demonstrate the effectiveness of integrating OCR and NLP, a clear research gap remains. Previous work often focuses on highly structured commercial documents like invoices [3] or applications in other specific domains such as resume screening [11], which have more predictable layouts compared to the semi-structured nature of administrative documents in community organizations. Furthermore, while some studies address the Indonesian context [10], they often do not offer an end-to-end management system that includes a practical user interface and a usability evaluation. This study addresses this gap by developing and validating a complete, end-to-end system tailored for the semi-structured and domain-specific nature of church administrative documents, utilizing a fine-tuned Indonesian language model, and measuring its real-world impact on user efficiency.

The advent of Transformer-based models [13], most notably BERT [14], has revolutionized the NLP landscape. BERT's bidirectional nature allows it to understand context far more effectively than previous models, leading to significant performance gains in tasks like NER. For languages other than English, language-specific models are crucial. Our work leverages IndoBERT, a BERT model specifically pre-trained on a large Indonesian corpus, which has been shown to outperform multilingual models on Indonesian NLP tasks [1]. By fine-tuning IndoBERT, we adapt a powerful, state-of-the-art architecture to the specific entities and document types relevant to our case study. The development of models like IndoBERT is part of a broader effort to advance Indonesian NLP, supported by benchmark datasets such as IndoNLU, which provide standardized evaluation for various language understanding tasks [15].

III. METHODOLOGY

Our methodology outlines an end-to-end pipeline that transforms an unstructured document image or PDF into structured, categorized information. The overall system architecture is visualized in Figure 1. This section details the dataset construction, preprocessing steps, model architecture, system deployment, and evaluation metrics.

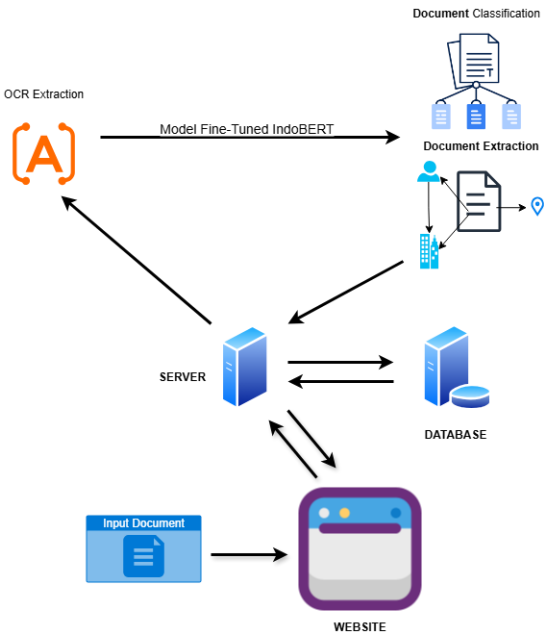


Figure 1. System architecture design

A. Dataset and Document Characteristics

The dataset was constructed from the administrative document archives of the St. Yohanes Rasul Chapel, covering documents published between 2024 and 2025. These documents, comprising permit letters (*Surat Perizinan*), invitations (*Surat Undangan*), and loan agreements (*Surat Peminjaman*), exhibit several key characteristics that pose challenges for automated processing:

- **Semi-structured Format:** They combine formal elements like letterheads with varied and often unpredictable layouts for the main content, including informal tables.
- **Domain-Specific Terminology:** The documents utilize specific language related to liturgical events, church roles (e.g., *Ketua Wilayah*), and unique asset names, which requires a model with domain-specific contextual understanding.
- **Visual Noise:** Many documents contain non-textual elements such as colored stamps, logos, and signatures that can interfere with the OCR process.

The initial collection consisted of 112 original documents. To mitigate the risks of overfitting due to a small dataset and to enhance model robustness, data augmentation was performed based on existing document templates, expanding the total dataset to 330 documents. This final dataset was then split into training (70%), validation (15%), and testing (15%) sets. To account for slight class imbalance, a stratified splitting strategy was employed to ensure that the proportion of each document category was maintained across all three sets. The detailed split is shown in Tabel I.

TABLE I. DATASET SPLIT

	Loan	Permit	Invitation
TRAIN	86	74	74
VALIDATION	21	17	11
TEST	16	16	16

B. Data Preprocessing and OCR

The process begins when a user uploads a document. The system first validates the file format (PDF, JPG, PNG). PDF files are converted into high-resolution images. Each image then undergoes a series of cleaning steps to optimize it for OCR, including the removal of non-textual elements like colored signatures and stamps using color segmentation, and conversion to grayscale. Finally, the cleaned image is fed into the PyTesseract engine, which performs OCR to extract the raw text content.

C. Classification and NER Model

The extracted text is processed by a fine-tuned IndoBERT model (Indolem/indobert-base-uncased). The model is adapted for a multi-task objective using two distinct output heads:

1. **Sequence Classification Head:** Attached to the [CLS] token's output, this head uses a fully connected layer to predict the document's category (Permit, Invitation, or Loan).
2. **Token Classification Head:** Applied to every token in the sequence, this head predicts an entity label based on the BIO (Beginning, Inside, Outside) scheme, enabling the extraction of specific information like dates, names, and locations.

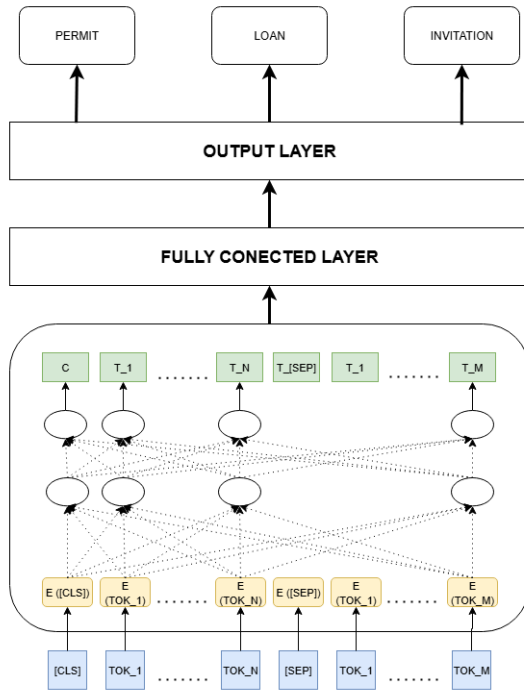


Figure 2. BERT Fine-Tuning Architecture for Classification

D. System Architecture and Deployment

To ensure a clear separation of concerns, the system is implemented using a service-oriented architecture, as depicted in Figure 3. This architecture was chosen to decouple the user-facing application from the computationally intensive machine learning tasks, allowing for independent scaling and maintenance.

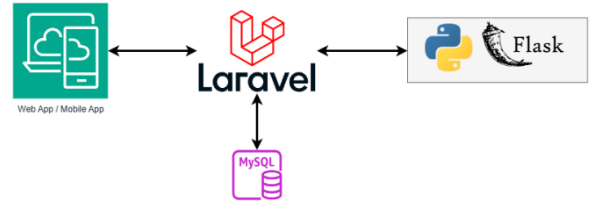


Figure 3. Service-Based System Architecture

- **Laravel Backend-for-Frontend (BFF):** The main application is built using the Laravel framework, which acts as the control center. It was selected for its robust features in handling user interactions, business logic, authentication, and secure data persistence through interactions with the MySQL database. It is the sole client authorized to communicate with the machine learning API.
- **Flask API for Machine Learning:** A dedicated microservice built with Python and Flask is responsible for all computational tasks. This approach was chosen to leverage Python's extensive ecosystem for NLP and computer vision. This service receives document files from the Laravel application, executes the entire pipeline including OCR with PyTesseract and inference with the fine-tuned IndoBERT model, and returns the structured results in JSON format.
- **MySQL Database:** This component serves as the persistent data storage layer, managed exclusively by the Laravel application to ensure data integrity.

E. Evaluation Metrics

To quantitatively assess the system's performance, the following metrics were used:

- **Document Classification:** The model's ability to correctly categorize documents was measured using Accuracy, calculated as the ratio of correctly classified documents to the total number of documents.
- **Named Entity Recognition (NER):** The NER task was evaluated using standard token-level metrics such as Precision, Recall, and F1-Score. The final reported F1-Score is a macro-average across all entity classes, which calculates the metric independently for each class and then takes the average, thereby treating all classes as equally important.

IV. EXPERIMENTS AND RESULTS

The fully integrated system was evaluated on a final hold-out set of 47 new, unseen documents to simulate real-world usage. The quantitative results for classification, NER, and usability are presented and discussed below.

A. Document Classification Performance

The classification model demonstrated high robustness, achieving an accuracy of 93.6%, correctly identifying 44 out of the 47 document types. An analysis of the confusion matrix in Figure 4 reveals the model's strengths and limitations. The model successfully classified all 15 PERMIT documents without error, indicating its strong ability to recognize the distinct features of this category.

However, three misclassifications were identified, stemming from significant semantic overlap in the document

content. Specifically, one LOAN document was mistaken for a PERMIT, another for an INVITATION, and one INVITATION document was misclassified as a PERMIT. This confusion arises from several factors:

- **Format Similarity:** The general structure and formatting of an INVITATION can be nearly identical to that of a PERMIT letter.
- **Content Overlap:** LOAN documents often contain detailed descriptions of the event for which the items are needed, giving them textual features similar to an INVITATION. Conversely, it is not uncommon for a PERMIT request to include a list of borrowed items, causing the model to confuse it with a LOAN document

This highlights that while the model is highly effective, its performance can be challenged by documents where category-defining keywords and structures are not mutually exclusive. Future work could address this by incorporating layout-aware features or developing a hierarchical classification approach to better navigate these nuanced cases.

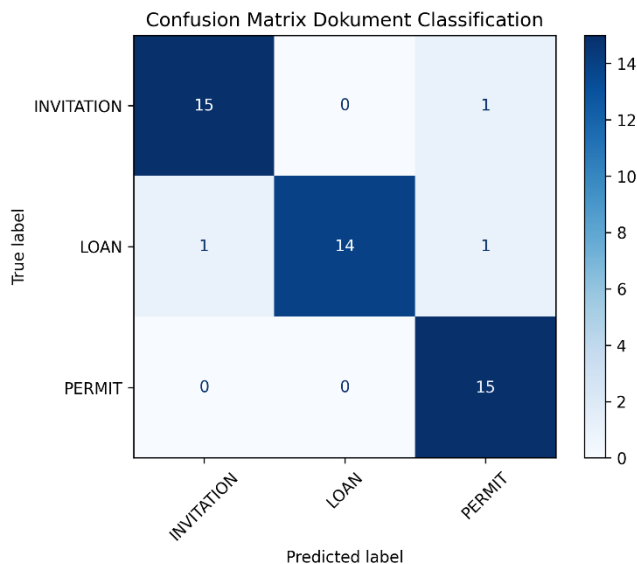


Figure 4. Confusion Matrix for Document Classification

B. Named Entity Recognition (NER) Performance:

In testing on the hold-out set, the fine-tuned NER model achieved an excellent macro F1-score of 0.989, proving its strong capability in identifying entities when provided with accurate text. However, a key finding from our end-to-end system evaluation is that the practical quality of NER is highly dependent on the initial OCR accuracy.

TABLE II. ENTITY PERFORMANCE

Entity	Precision	Recall	F1-Score	Quantity
SCHEDULE_TIME	1.0	1.0	1.0	14
DOCDATE	1.0	1.0	1.0	47
EMAIL	1.0	1.0	1.0	16
CITY	1.0	1.0	1.0	27
PHONE	1.0	1.0	1.0	10
ITEMQTY	0.99	0.99	0.99	137
SUBJECT	1.0	0.97	0.98	46

ITEM	0.98	0.97	0.97	144
SCHEDULE_ITEM	0.96	0.96	0.96	32
EVTDATE	1.0	0.90	0.94	51
PEOQTY	1.0	0.90	0.94	10
POSITION	0.98	0.90	0.94	87
EVTTIME	0.97	0.90	0.93	50
EVT	0.94	0.92	0.93	51
PER	0.94	0.90	0.92	98
EVTLOC	0.95	0.88	0.91	50
ORG	0.84	0.97	0.90	91
RECIPIENT	0.88	0.91	0.90	61
DOCNUM	0.75	0.75	0.75	33
SCHEDULE_DURATION	0	0	0	7

Tabel II provides a detailed breakdown of the model's performance for each entity. The model demonstrated exceptional capability in recognizing highly structured entities. As shown, entities such as SCHEDULE_TIME, DOCDATE, EMAIL, CITY, and PHONE all achieved perfect F1-scores of 1.0, indicating the model's robustness in extracting factual, unambiguous information. Conversely, the performance on the DOCNUM entity was significantly lower (F1-score 0.758). Investigation revealed that this was primarily due to OCR errors, where the OCR engine struggled to correctly transcribe characters such as Roman numerals (e.g., "VIII" being incorrectly read as "V I I I" or "VIII"). This highlights that while the core NLP model is powerful, the overall system performance is a cumulative result of its entire pipeline, with the OCR stage acting as a potential bottleneck.

C. System Usability and Efficiency

To measure the system's practical impact, we evaluated its effect on the document management workflow. A comparative time analysis was conducted, and the results, as summarized in Tabel III, show a significant reduction in processing time.

TABLE III. TIME COMPARISON OF DOCUMENT MANAGEMENT WORKFLOWS

Process	Average Time per Document (seconds)
Manual	89
Using the system	27

The data in Tabel III reveals that the system reduced the average document processing time from 89 seconds to just 27 seconds. This constitutes a 69.6% improvement in efficiency, a substantial enhancement that validates the system's effectiveness in accelerating administrative tasks.

Furthermore, user acceptance testing was conducted with the primary user (the chapel secretary), yielding a System Usability Scale (SUS) score of 77.5. According to standard interpretation scales, a score in this range indicates "Good" usability. This result, combined with the quantitative efficiency gains, confirms that the system is not only functionally powerful but also well-received and easy to use in a real-world context.

D. System Implementation and User Interface

To validate the system's practical application, a web-based user interface was developed. This interface allows users (the chapel administrators) to upload documents and review the results of the automated analysis. Figure 5 displays the system's output screen after processing a document.

The interface is divided into two main panels. On the right, the original document is displayed for reference. On the left, the structured information extracted by the system is presented in a clear and editable format. This includes:

- **AI Analysis Result:** The system correctly identifies the document type (e.g., "peminjaman" or loan agreement) and extracts key information such as the event name ("serah terima jabatan pengurus omk"), date, and location.
- **Structured Data Fields:** Specific details, including the list of borrowed items and their quantities, are neatly organized for quick verification.
- **User Verification and Control:** The interface provides "Edit" and "Submit" buttons, giving the user full control to verify, correct any potential extraction errors, and save the final, accurate data into the database.

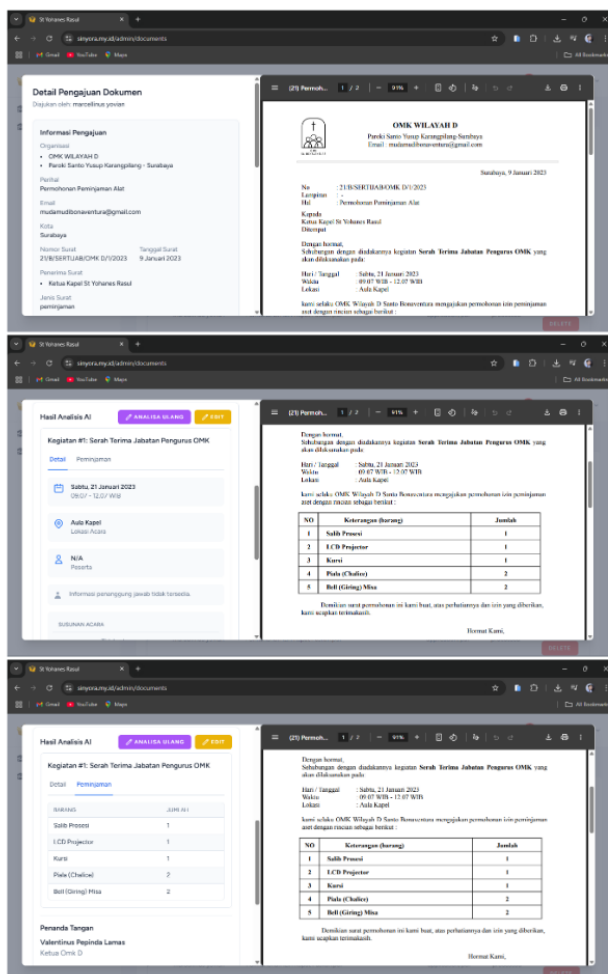


Figure 5. System User Interface Displaying Analysis Results.

V. CONCLUSION

This study successfully demonstrated the feasibility and effectiveness of an integrated system using OCR and a fine-tuned BERT model for automating administrative document management. The system is capable of accurately classifying documents, achieving 93.6% accuracy on new, unseen data, and extracting key information with the NER model reaching an F1-Score of 0.989 on the test set. This automation leads to significant, measurable efficiency gains, reducing manual processing time by up to 69.6%. Furthermore, the system demonstrates strong user acceptance, achieving a System Usability Scale (SUS) score of **77.5**, which indicates 'Good' usability.

Despite these promising results, this study has several limitations that must be acknowledged. The primary limitation is the reliance on a dataset from a single institution, which is relatively small even after augmentation. This raises concerns about the model's generalizability to documents from other organizations with different templates and writing styles. Moreover, the small dataset size introduces a potential risk of overfitting, where the model may have memorized specific patterns in the training data rather than learning generalizable features. Although the high performance on the hold-out set suggests good generalization, this risk cannot be fully dismissed. Another key limitation is the direct impact of OCR quality on the downstream NLP model's performance, where errors in text extraction can create a ceiling for the overall system's accuracy.

Future work should focus on addressing these limitations. First, expanding the dataset with more varied documents is crucial to improve model generalization and mitigate overfitting. Second, exploring more advanced computer vision techniques for image preprocessing could enhance OCR accuracy on challenging, low-quality documents. Finally, refining the post-processing logic to better handle complex entity structures (e.g., multiple instances of the same entity type) would improve the granularity and accuracy of the final extracted data. Despite these limitations, this research provides a solid framework for applying modern NLP to solve practical administrative challenges in community organizations.

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